

ROS 2 guide

Linux installation

Open the Linux command terminal and paste the following commands (you will need the admin permissions for this installation)

<https://wiki.ros.org/noetic/Installation/Ubuntu>

Link to the working and confirmed installation process

Windows installation

The fastest way for windows is to install the wsl (windows linux system) virtual machine and do the same path as for Linux installation.

IMPORTANT!!! Make sure you will use the Ubuntu 20.04 ROS 2 is confirmed to work with this distribution other versions may not work properly.

To install open powershell terminal as admin and type:

```
wsl.exe --list --online
```

```
wsl.exe --install Ubuntu-20.04
```

And to run it: `wsl.exe -d Ubuntu-20.04`

<https://wiki.ros.org/noetic/Installation/Ubuntu>

Link to the working and confirmed installation process

ROS setup - run roscore

To run ros system you will need separate terminals.

Open a terminal and run roscore as hub for ros system. You should get something like this:

-> **roscore**

```
Checking log directory for disk usage. This may take a while.
Press Ctrl-C to interrupt
Done checking log file disk usage. Usage is <1GB.

started roslaunch server http://LukaszK:40181/
ros_comm version 1.17.4

SUMMARY
=====

PARAMETERS
* /roscdistro: noetic
* /rosversion: 1.17.4

NODES

auto-starting new master
process[master]: started with pid [1073]
ROS_MASTER_URI=http://LukaszK:11311/

setting /run_id to 568195d8-ec8a-11f0-b479-4f532171979b
process[rosout-1]: started with pid [1083]
started core service [/rosout]
|
```

Device port attach to virtual machine (Windows ONLY)

If you used the ubuntu virtual machine you will need to attach a COM port to in order to let the ROS system communicate with the device. You can do it in powershell(**as admin**) by entering commands:

usbipd list -> this will list all the ports in your system find the device port and remember its BUSID. It should look like this:

```
PS C:\WINDOWS\system32> usbipd list
Connected:
BUSID  VID:PID  DEVICE                                STATE
1-2    046d:c534 USB Input Device                      Shared
1-3    17ef:c000 Zephyr FOR Zhanlu, USB Input Device Not shared
1-4    1a86:7523 USB-SERIAL CH340 (COM10)              Not shared
1-5    0bda:5a13 Smart-Antenna Controller, USB DFU Device Not shared
1-6    048d:c990 USB Input Device                      Not shared
1-12   04f2:b7b6 Integrated Camera, Camera DFU Device, Tobii Experience Not shared
1-13   27c6:659a Goodix MOC Fingerprint                 Not shared
1-14   8087:0033 Intel(R) Wireless Bluetooth(R)        Not shared
```

usbipd bind --busid 1-4

usbipd attach --wsl --busid 1-4 -> those two commands will attach the port (in the example busid 1-4 but replace it with the one you got on your PC) to the windows virtual machine now ROS can access the port for communication

ROS setup - establish a serial connection with a board

In a separate terminal run the command to establish a bridge to the ROS device (you will need the COM port number to type in the command)

Run the command:

```
roslaunch rosserial_arduino serial_node.py _port:=/dev/ttyUSB0 _baud:=57600
```

The `_port` parameter may differ. You should get the following output:

```
[INFO] [1767875213.210898]: ROS Serial Python Node
[INFO] [1767875213.213078]: Connecting to /dev/ttyUSB0 at 57600 baud
[INFO] [1767875215.325275]: Requesting topics...
[INFO] [1767875215.352094]: Note: subscribe buffer size is 280 bytes
[INFO] [1767875215.353580]: Setup subscriber on servo [std_msgs/Int16MultiArray]
```

Run ROS commands

Open the 3rd terminal in the wls and run the commands for the ROS subscription system the example one:

```
rostopic pub servo std_msgs/Int16MultiArray "{layout: {dim: [{label: '', size: 0, stride: 0}], data_offset: 0}, data: [50, 60]}" --once
```

The data parameters in that command are the servo positions of the arm. After sending the command you should get the confirmation and observe the movement of the arm.

```
lukko@LukaszK:/mnt/c/Users/Admin$ rostopic pub servo std_msgs/Int16MultiArray "{layout: {dim: [{label: '', size: 0, stride: 0}], data_offset: 0}, data: [50, 60]}" --once
publishing and latching message for 3.0 seconds
```

The whole setup

```

Invalid command line argument: -system
Please use 'ml.exe --help' to get a list of supported arguments.
PC @ C:\Users\Administrator\OneDrive\Documents> cd /d C:\Users\Administrator\OneDrive\Documents
C:\Users\Administrator\OneDrive\Documents> cd /d C:\Users\Administrator\OneDrive\Documents
... Logging to /home/luksasz/.ros/log/669819568-e8ba-11f0-b479-43521719799b/roslaunch-1063-Log
Checking log directory for disk usage. This may take a while.
Press Ctrl-C to interrupt.
Done checking log directory disk usage. Usage is <GB>.

started roslaunch server http://10.0.2.15:1131/
ps: core version 1.37.4

```

MAIN ROSCORE TERMINAL

```
SUMMARY
=====

PARAMETERS
 * /rostdistro: noetic
 * /rosversion: 1.17.4

NODES
```

```
auto-starting new master
process[master]: started with pid [1873]
ROS_MASTER_URI=http://LukaszK:11311/

setting /run_id to 568195d8-ec8a-11f8-b4
process[rosout-1]: started with pid [108]
started core service [/rosout]
```

```
During handling of the above exception, another exception occurred:
```

```
Traceback (most recent call last):
  File "/opt/ros/noetic/lib/python3/dist-packages/roscpp_serial_client.py", line 374, in _init_
    self.port = Serial(port, baud, timeout=self.timeout, write_timeout=10)
  File "/usr/lib/python3/dist-packages/serial/serialutil.py", line 206, in _init_
    self.open()
  File "/usr/lib/python3/dist-packages/serial/serialposix.py", line 269, in open
    raise SerialException(msg_err, "could not open port (%s): (%s)" % (self.port, msg))
serial.SerialException: [Error 2] could not open port (/dev/ttyUSB0): [Error 2] No such file or directory
vty/vty$
```

```

During handling of the above exception, another exception occurred:

Traceback (most recent call last):
  File "/opt/ros/noetic/lib/python3/dist-packages/rospy/core.py", line 598, in signal_shutdown
    NO
  File "/opt/ros/noetic/lib/python3/dist-packages/rospiral/serialClient.py", line 354, in shutdown
    self.closeConnection()
  File "/opt/ros/noetic/lib/python3/dist-packages/rospiral/serialClient.py", line 422, in closeConnection
    self.port.close()
AttributeError: 'serialClient' object has no attribute 'port'
[rospiral@rosmaster ~]$ rosrun rospiral rospiral_serial_node.py --port=/dev/ttyUSB0 --baud=57600
[INFO] [1767876231.216868]: ROS Serial Node
[INFO] [1767876231.219782]: Connecting to /dev/ttyUSB0 at 57600 baud
[INFO] [1767876231.325278]: Requesting topic...
[INFO] [1767876231.325906]: Notes subscriber buffer size is 280 bytes
[INFO] [1767876231.353588]: Start subscriber on servo [std_msgs/Int16Array]

```

COM port bridge terminal



```
lukasz@lukaszC:/mnt/c/Use...
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows
```

```
PS C:\Users\Admin> wsl.exe Ubuntu-20.04
luk00z@lukasztk:/mnt/c/Users/Admin$ rostopic pub std_msgs/Int64MultiArray "[{layout: [{dim: [{label: '', size: 0, str
ide: 0}], data_offset: 0}, data: [50, 60]}]" --once
publishing and latching message for 3.0 seconds
luk00z@lukasztk:/mnt/c/Users/Admin$
```

```

2643f0c2-87ca-4746-aed-f3c49eaeaf    USB Serial Device (COM),   USB 374d serial debug unit
3125f3c3-125c-4706-8cc7ecddcd      Silicon Labs CP210x to UART Bridge (COM)
2643f0c2-87ca-4746-930d-3350da285ff8 USB Serial Converter

PS C:\WINDOWS\system32> wslinfo build --build-id -l
ConnectId          BuildID            DEVICE                                     STATE
-----
1                   46dd-4334         USB Input Device                          Shared
2                   17ef-F0D0        Zephyr for Xnomo, US Input Device       Not shared
3                   4ad6-C0A0        USB-SERIAL CHANL (CON1)                  Not shared
5                   Wnda-SB1S        Smart Antenna Controller, USB GPU Device Not shared
6                   4ad6-C0A0        USB Input Device                         Not shared
7                   08af-F7B0        Integrated Camera, CPU & GPU Device, Tobi Experience Not shared
11                  86dc-dc     No fingerprint                           Not shared
12                  86d0-B03F        Intel(R) Wireless Bluetooth(R)           Not shared

Persisted:
Device             BUILD ID
-----
2643f0c2-87ca-4746-aed-f3c49eaeaf    USB Serial Device (COM),   USB 374d serial debug unit
3125f3c3-125c-4706-8cc7ecddcd      Silicon Labs CP210x to UART Bridge (COM)
2643f0c2-87ca-4746-930d-3350da285ff8 USB Serial Converter

PS C:\WINDOWS\system32> wslinfo attach --attach -l -b
wslinfo info: wslinfo distribution "Ubuntu" does not support attaching the device via available in all wsl distributions.
wslinfo info: loading local host module...
wslinfo info: detected networking mode "nat".
PS C:\WINDOWS\system32> wslinfo ip address 17.11.01.1 to reach the host.
```

PORT ATTACH TERMINAL (WINDOWS ONLY)